

Stats lec

① Hypothesis testing. ✓

② P-value ✓

③ t-test ✓

④ z-test

⑤ Chi-squares test

① Hypothesis testing :-

Population :-

Sample :-

✓ Null Hypothesis (H_0) :- Default assumptions.

✓ Alternate Hypothesis (H_1) :- What we want to prove.

✓ Significance level (α) :- Risk we are willing to take of being wrong.

✓ Test stats :- (z, t, χ^2)

P-value :-

- ①
- ✓ H_0 (Null Hypothesis) :- The accused is innocent
 - ✓ H_1 (Alternate Hypothesis) :- The accused is guilty. $\checkmark$

② Choose Significance level (α)

$$\alpha = 0.05$$

③ Choose Right test :-

z-test :- large sample or known Population SD.

$\checkmark$ t-test :- Small sample (< 30) and unknown SD.

* Chi-Square (χ^2) :- Categorical data.

* ANOVA $\rightarrow$ > 2 group comparison.

④ Compute Test Stat

$\rightarrow \chi^2, \text{ANOVA}$

⑤ Compute P-value $\approx$

⑥ Compare P-value with α

$p < \alpha \rightarrow$ Reject H_0

$p \geq \alpha \rightarrow$ Fail to reject H_0

Claim from teacher is that on an average my students get 70 in math.

records of student math score $\rightarrow$ [65, 68, 75, 80, 72]

① ✓

$$H_0: \mu = 70$$
$$H_1: \mu \neq 70$$

✓ [5, 10, 4, 2, 6] $\leftarrow \bar{x}$
 μ

② ✓

$$\alpha = 0.05$$

③ ✓ I will be doing t-test as (sample < 30)

④ ✓ Compute + :-

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$\bar{x} \Rightarrow$ sample mean

$\mu \Rightarrow$ Population mean = 70

$s \Rightarrow$ Sample SD

$\sqrt{n} \Rightarrow$ Sample number of data

$$\bar{x} = 72$$

$$\mu = 70$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$n = 5$$

$$t = 0.76$$

$$t =$$

$$= 5.87$$

⑤ Compute P-value :-

$$n = 5$$

$$\text{d.f.} = n - 1 = 5 - 1 = 4$$

$$t = 0.76$$

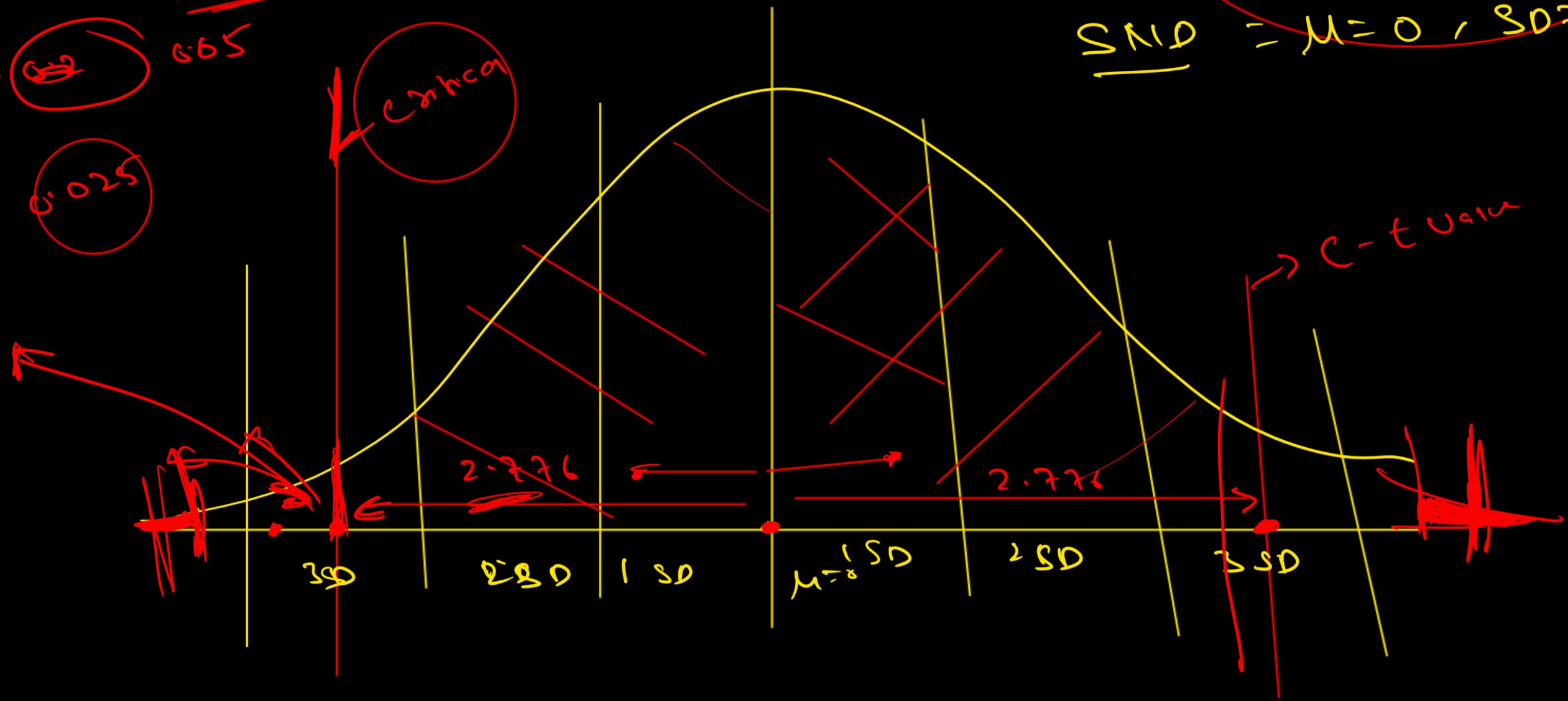
2.9

Critical t-value for $\alpha(0.05)$ (two-tailed) $\approx \pm 2.776$

$$\alpha = \text{GBS}$$

SND = $\mu = 0$, $SD = 1$

$$\frac{\sigma}{2} = 0.025$$



$t = 3.75$
 $C-t = \pm 2.276$

$H_0 \rightarrow$ rejected

